

1 diffblas: algorithmically differentiated BLAS routines

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6 Summary

7 diffblas is a library that provides algorithmically differentiated ([Griewank & Walther, 2008](#))
8 BLAS routines from their reference implementations in [LAPACK](#) on GitHub using the automatic
9 differentiation tool Tapenade ([Hascoët & Pascual, 2013](#)). It supports four modes: forward
10 (`_d`), vector forward (`_dv`), reverse (`_b`), and vector reverse (`_bv`).

11 In addition to differentiating the standard Fortran-style BLAS interface, diffblas also provides
12 differentiated CBLAS routines, facilitating interoperability with C and other languages. Its API
13 mirrors BLAS/CBLAS, with additional arguments specifying differentiation variables, making
14 it straightforward to integrate into existing workflows.

15 diffblas calls the underlying standard BLAS implementation, and is agnostic to the backend
16 (OpenBLAS, BLIS, MKL, Apple Accelerate), ensuring both performance and portability.

17 By providing accurate and efficient derivatives of linear algebra operations, diffblas facilitates
18 gradient-based optimization, sensitivity analysis, and a wide range of scientific computing
19 applications

20 Statement of need

21 Linear algebra routines such as those in LAPACK are fundamental to scientific computing,
22 optimization, and machine learning. However, they do not provide derivatives, which are often
23 required for gradient-based algorithms. Current approaches either rely on hand-coded derivatives
24 or generic automatic differentiation applied to high-level code, which can be inefficient or
25 error-prone. diffblas addresses this gap by providing algorithmically differentiated BLAS
26 routines directly from reference LAPACK implementations and following relevant differentiation
27 rules ([Giles, 2008](#)). This enables accurate and efficient computation of derivatives while
28 preserving compatibility with existing BLAS-based codes.

29 State of the field

30 Automatic differentiation (AD) tools such as Tapenade ([Hascoët & Pascual, 2013](#)), ADOL-C
31 ([Griewank et al., 1996](#)), or Taf provide general mechanisms to compute derivatives of code.
32 However, applying AD naively to low-level BLAS or LAPACK routines can be inefficient due to
33 loops, memory layout, and caching issues ([Jonasson et al., 2020](#)). Specialized libraries like
34 diffblas that generate differentiated routines directly from reference implementations combine
35 the reliability of LAPACK with the efficiency of AD, bridging a gap in current scientific
36 computing workflows.

Research impact statement

This work was inspired in part by a need to differentiate a Fortran code (Hascoët et al., 2026) that uses BLAS and LAPACK routines, and to use the differentiated application for gradient-based optimization.

Providing both the standard and CBLAS interfaces ensures that diffblas can be adopted across different programming environments, facilitating derivative computations in diverse scientific computing projects. Precompiled artifacts on GitHub further simplify integration, enabling rapid deployment in multiple languages and scientific computing projects.

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AI usage disclosure

Generative AI was used to ... ChatGPT was used to check spelling, grammar, and clarity of the English text in this paper.

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